

Engineering Reboot 2026

Week-02 Engineering Lab Workbook v3.2

Self-contained execution workbook. Designed to minimize context switching and interpretation.

Day 06 – Conditional Logic

Session Budget (60 Minutes)

Theory Review – 10 min

Exercise Work – 35 min

Notes – 5 min

Journal – 5 min

Git Commit/Push – 5 min

Exercise 01 – Age Classifier

Filename: age_classifier.py

Prompt: Input age:

Rules: 0-12 Child, 13-19 Teen, 20-64 Adult, 65+ Senior

Test Case

Input age: 5

Classification: Child

Test Case

Input age: 13

Classification: Teen

Test Case

Input age: 35

Classification: Adult

Test Case

Input age: 70

Classification: Senior

Screenshots:

AgeClassifier-Code.png

AgeClassifier-Test-T01.png

AgeClassifier-Test-T02.png

AgeClassifier-Test-T03.png

AgeClassifier-Test-T04.png

Exercise 02 – Temperature Advisor

Prompt: Input temperature in Fahrenheit:

Below 50 = Wear Jacket

50-75 = Comfortable

Above 75 = Warm Weather

Directory Structure

Day-XX/

Program File

Screenshots/

Notes/

Journal/

Notes Template

Concepts Learned

Observations

Questions

Key Takeaways

Journal Template

Time Started

Time Finished

What Went Well

What Was Difficult

Improvement For Tomorrow

Git Commands

git status

git add .

git commit -m 'Day Complete'

git push

Acceptance Criteria

All exercises complete.

All screenshots captured.

Notes complete.

Journal complete.

Git push complete.

Day 07 – Boolean Logic

Session Budget (60 Minutes)

Theory Review – 10 min

Exercise Work – 35 min

Notes – 5 min

Journal – 5 min

Git Commit/Push – 5 min

Exercise 01 – Voting Eligibility Checker

Filename: eligibility_checker.py

Prompt 1: Input age:

Prompt 2: Citizen (Y/N):

Input age: 25

Citizen: Y

Eligible: YES

Input age: 16

Citizen: Y

Eligible: NO

Exercise 02 – Building Access Control

Prompt for badge_present and manager_override.

Access granted if either valid badge OR manager override is true.

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Time Started

Time Finished

What Went Well

What Was Difficult

Improvement For Tomorrow

Git Commands

git status

git add .

git commit -m 'Day Complete'

git push

Acceptance Criteria

All exercises complete.

All screenshots captured.

Notes complete.

Journal complete.

Git push complete.

Day 08 – Loops

Session Budget (60 Minutes)

Theory Review – 10 min

Exercise Work – 35 min

Notes – 5 min

Journal – 5 min

Git Commit/Push – 5 min

Exercise 01 – Counter Program

Display numbers 1 through 20.

Expected display includes every number on its own line.

Exercise 02 – Even Number Generator

Display even numbers 2 through 20.

Expected display: 2 4 6 8 10 12 14 16 18 20

Exercise 03 – Sum Calculator

Calculate sum of numbers 1 through 100.

Expected result: 5050

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Time Finished

What Went Well

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Git Commands

```
git status
git add .
git commit -m 'Day Complete'
git push
```

Acceptance Criteria

All exercises complete.
All screenshots captured.
Notes complete.
Journal complete.
Git push complete.

Day 09 – Loop Practice

Session Budget (60 Minutes)

Theory Review – 10 min

Exercise Work – 35 min

Notes – 5 min

Journal – 5 min

Git Commit/Push – 5 min

Exercise 01 – Square Pattern

Input size: 5

Display 5x5 star pattern.

Exercise 02 – Right Triangle

Rows: 5

Expected pattern grows one star per row.

Exercise 03 – Inverted Triangle

Rows: 5

Expected pattern decreases one star per row.

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Time Started

Time Finished

What Went Well

What Was Difficult

Improvement For Tomorrow

Git Commands

git status
git add .
git commit -m 'Day Complete'
git push

Acceptance Criteria

All exercises complete.
All screenshots captured.
Notes complete.
Journal complete.
Git push complete.

Day 10 – Mini Project

Session Budget (60 Minutes)

Theory Review – 10 min

Exercise Work – 35 min

Notes – 5 min

Journal – 5 min

Git Commit/Push – 5 min

Mini Project – Menu Driven Calculator

Filename: menu_driven_calculator.py

Menu Options:

1 Add

2 Subtract

3 Multiply

4 Divide

5 Exit

Required Test Matrix (minimum):

$2+3=5$

$10-4=6$

$5*7=35$

$20/5=4$

Division by zero handling

Exit option handling

Directory Structure

Day-XX/

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Notes/

Journal/

Notes Template

Concepts Learned

Observations

Questions

Key Takeaways

Journal Template

Time Started

Time Finished

What Went Well

What Was Difficult

Improvement For Tomorrow

Git Commands

git status

git add .

git commit -m 'Day Complete'

git push

Acceptance Criteria

All exercises complete.

All screenshots captured.

Notes complete.

Journal complete.

Git push complete.